

# Levi Mackay

Computer Science Student, Open Source and Backend Systems

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Computer science student at BYU-Idaho, 3.98 GPA, graduating December 2027 and looking for a Summer 2027 software engineering internship. Maintains an open source AI coding agent with merged pull requests from developers I have never met, and ships production web work that local businesses pay for.

## OPEN SOURCE

### Lydia, a local AI coding agent

github.com/levimackay/lydia-cli · MIT · 36 stars, 8 forks

- A terminal coding agent that reads, edits, and tests code entirely against local Ollama models. No API keys, no subscription, and nothing leaves the machine. Roughly 6,500 lines of Python across 59 modules.
- 376 tests across 51 files run in GitHub Actions on every push, and the suite has to be green before anything merges.
- Two developers found the project on their own and opened pull requests that are now merged: Linux systemd user service support, and a fix for boolean config values stored as strings.

## EXPERIENCE

### Main Street Sites, Founder and Developer · BYU-Idaho Sandbox Incubator

Rexburg, ID · May 2026 to Present

- Building a self-serve web design studio for local businesses on Next.js 16 and TypeScript, with Neon Postgres behind Drizzle ORM, Clerk for auth, and Stripe for recurring billing.
- Shipped production redesigns for two Rexburg businesses, and ran the outreach and sales pipeline end to end rather than handing it off.

### BYU-Idaho, Computer Science Teaching Assistant

Rexburg, ID · Apr 2026 to Present

- Work with students on programming fundamentals and debugging strategy, and grade assignments with written feedback rather than a score alone.
- Coach algorithmic thinking one on one, mostly by asking the next question instead of handing over the answer.

### BYU-Idaho, CSE Student Employability and Outreach Assistant

Rexburg, ID · Apr 2026 to Present

Work with CS faculty on internship promotion, employer visits, and technical workshops, and track engagement numbers to find what actually brings students in.

### SafeStreets, Dealer Operations and IT Specialist

Rexburg, ID · Mar 2025 to Aug 2025

Managed dealer account data, permissions, and system access, and resolved record discrepancies several teams depended on.

## PROJECTS

### RexNest, off campus housing marketplace

Next.js 16, Supabase Postgres, Stripe

- Ranks apartments with a weighted scoring function over population normalized price, distance, and amenities, plus Bayesian style shrinkage on reviews so two glowing ones do not outrank fifty good ones.
- 13 Postgres tables behind 129 lines of row level security policy, called under security definer with a pinned search path. 114 tests across 10 files.
- Monetization built on Stripe: recurring subscriptions for property managers and one time paid boosts for listings.

### Serbo-Croatian dictionary, digitization and typesetting pipeline

Python, Typst

- Turned a 208 page scanned dictionary and two years of handwritten field notes, collected while living in Croatia, Serbia, and Bosnia, into a typeset 301 page reference of 5,709 entries.
- About 1,000 lines of Python across a 7 stage pipeline, from extraction and merging through reverse index generation and Typst assembly. Every field the pipeline adds or corrects is flagged in the output instead of quietly overwriting the source.

## EDUCATION

### Brigham Young University Idaho

Sep 2024 to Dec 2027 (expected)

B.S. Computer Science · 3.98 GPA

## SKILLS

**Languages:** Python, TypeScript, JavaScript, SQL, C# · **Web:** Next.js, React, Tailwind, three.js

**Data:** PostgreSQL, Supabase, Drizzle ORM, row level security, schema design · **Tooling:** Git, GitHub Actions, pytest, Stripe, Ollama